WEEKLY PROGRESS REPORT

WEEK 03: IoT ARCHITECTURE, DESIGN & DEPLOYMENT CHALLENGES

Intern Name:	Gayatri	**Week Ending:**	Week 03 (September 30, 2026)
Intern Domain:	Embedded Systems & IoT	**Selected Project:**	IoT Architecture & Deployment Study

IoT LAYERS	DEPLOYMENT	CLOUD & CASE STUDIES
Devices, sensors, connectivity & gateways	Reliability, power, security & scalability	Cloud platforms and practical IoT applications

I. OVERVIEW

This week focused on understanding the structure of an Internet of Things (IoT) solution and the practical decisions required to move from an IoT concept toward a deployable system. The learning was aligned with the Embedded Systems and IoT internship curriculum, especially IoT architecture, sensors, physical devices, platform design, cloud systems and case studies.

The main activity was to study IoT as an end-to-end system. I analysed how sensing devices generate information, how connectivity and gateways transport or preprocess it, how cloud platforms can store and process it, and how applications turn the resulting information into useful services.

I also studied deployment challenges including sensor reliability, power management, connectivity, scalability, security, data handling, environmental conditions, maintenance and cost. These topics helped connect theoretical architecture with real-world requirements.

II. ACHIEVEMENTS

1. Understanding IoT Layered Architecture

I studied the major components of an IoT architecture: physical devices and sensors, connectivity/networking, gateways or edge components, cloud/processing services and applications. Each component has a distinct role, but the complete system must be designed as one connected solution.

Component	Role Studied
Sensors & Physical Devices	Measure physical parameters and interact with the physical environment.
Connectivity	Transfers device data through an appropriate communication/network technology.
Gateway / Edge	Connects devices to wider networks and can aggregate or preprocess information.
IoT Cloud Platform	Supports data ingestion, storage, processing and device/service management.
Application Layer	Presents data through dashboards, alerts, reports or automated services.

2. Sensor Selection

Sensor selection was studied as a design decision. Important factors include the measured parameter, range, accuracy, resolution, response time, power consumption, operating environment, interface, reliability and cost.

3. Gateway Selection

Gateway selection depends on device count, supported protocols, processing needs, connectivity options, security requirements, power availability and scalability. The gateway can connect devices with wider networks and may aggregate or preprocess data.

4. Connectivity Selection

Connectivity was analysed in terms of range, bandwidth, latency, power consumption, reliability, availability and cost. Communication technology should be selected according to the application's requirements.

III. DESIGN CHALLENGES IN IoT LAYERS

5. IoT Cloud Platforms

I studied the role of IoT cloud platforms in receiving device data, storing information, processing events and supporting applications. Cloud infrastructure can provide scalable services, but the design still requires attention to security, data management, cost and scalability.

Challenge	What I Studied / Understood
Sensor reliability	Calibration drift, environmental effects, physical damage and inconsistent readings can affect long-term reliability.
Power management	Battery-powered devices need efficient sensing, processing and communication to extend operating life.
Connectivity	Real locations may have weak or intermittent connectivity, so the network must match the environment.
Scalability	A prototype architecture may need changes when the number of connected devices becomes large.
Security	Devices, communication paths, gateways and cloud services require appropriate protection and access control.
Data management	Continuous sensing can generate large data volumes, requiring suitable processing, storage and retention.
Environmental conditions	Temperature, humidity, dust, vibration and outdoor exposure can affect device performance.
Maintenance	Distributed devices make updates, fault detection, calibration and replacement more difficult.
Cost	Total cost includes hardware, gateways, connectivity, cloud services, installation and maintenance.

IV. REAL-WORLD IoT USE-CASE STUDY

The provided course material lists Home Intrusion Detection, Weather Monitoring System, Air Pollution Monitoring, Smart Irrigation and Energy Harvesting as IoT case-study areas. These examples were used to understand how common architectural principles can be adapted to different physical problems.

Case Study	System Focus	Important Design Point
Weather Monitoring	Environmental measurements collected through sensors.	Outdoor reliability and continuous connectivity.
Air Pollution Monitoring	Collection of air-quality information.	Sensor quality, calibration and coverage.
Smart Irrigation	Environmental/soil information used for irrigation decisions.	Power efficiency, field connectivity and timely control.
Home Intrusion Detection	Detection of security-related events.	Reliability, security and timely event handling.
Energy Harvesting	Monitoring energy generation or availability.	Low-power operation and efficient communication.

V. CURRICULUM CONNECTION

Curriculum Area	Week 03 Connection
Unit 1 – Introduction to Embedded System and IoT	IoT definition, sensors, applications, issues and challenges.
Unit 2 – Embedded IoT Platform Design Methodology	Requirement-oriented thinking and system-level design decisions.
Unit 3 – Pillars of Embedded IoT and Physical Devices	Physical devices, sensors, gateways and connected components.
Unit 4 – Web of Things and Cloud of Things	Cloud computing, middleware and cloud-based IoT architecture.
Unit 5 – IoT Cloud Offerings and Case Studies	Cloud services and real-world IoT applications.

VI. CHALLENGES AND HURDLES

1. Understanding the complete IoT flow: The main difficulty was connecting sensors, connectivity, gateways, cloud platforms and applications into one complete system. I addressed this by studying the architecture layer-by-layer and tracing how data moves through the system.

2. Prototype versus deployment: A small demonstration can prove sensing and communication, but a real deployment also requires reliability, power management, security, maintenance, scalability, environmental protection and cost considerations. I addressed this by analysing deployment challenges alongside architecture.

3. Selecting technologies according to requirements: Different IoT applications can require different sensors, gateways and connectivity methods. I compared selection factors such as range, power, accuracy, latency, reliability and cost rather than assuming one technology fits every application.

VII. LESSONS LEARNED

• IoT is an end-to-end system involving physical devices, communication, processing and applications.

• Sensor, gateway and connectivity choices must be based on the requirements of the intended application.

• Deployment introduces practical issues that may not appear in a small prototype.

• Power, reliability, security, scalability and maintenance need to be considered during architecture design.

• Cloud platforms provide useful infrastructure, but data management and cost still need planning.

• Real-world case studies help connect theoretical IoT architecture with practical applications.

VIII. LEARNING RESOURCES

1. Internship lecture material covering IoT layered architecture, sensor selection, gateway selection, connectivity selection, IoT cloud platforms and deployment challenges.

2. Embedded Systems and IoT course syllabus covering Units 1–5, including IoT architecture, platform design methodology, physical devices, Web/Cloud of Things and IoT case studies.

3. Adrian McEwen and Hakim Cassimally — Designing the Internet of Things, John Wiley and Sons Ltd., 2014.

4. Vijay Madisetti and Arshdeep Bahga — Internet of Things: A Hands-on Approach, Universities Press, 2015.

5. Dieter Uckelmann, Mark Harrison and Florian Michahelles — Architecting the Internet of Things, Springer, 2011.

IX. NEXT WEEK'S GOALS

• Continue Embedded IoT Platform Design Methodology, especially purpose/requirement, process, information model and service specification.

• Study physical IoT devices and Raspberry Pi interfaces/programming as listed in the curriculum.

• Select a practical IoT use case and map its sensor, gateway, connectivity, cloud and application requirements.

• Evaluate the proposed architecture for reliability, security, power consumption, scalability and cost.

• Maintain a structured weekly record of learning, challenges and progress.

X. ADDITIONAL COMMENTS

This week helped me understand that successful IoT development is not limited to connecting a sensor to the internet. A complete solution requires coordinated decisions across physical devices, communication, gateways, cloud services and applications. Studying deployment challenges highlighted the importance of designing for real-world conditions from the beginning.

Weekly Takeaway: Design the complete IoT system around the application's requirements, then evaluate every layer for reliability, connectivity, power, security, scalability and practical deployment.

XI. SUBMISSION CHECK

Deliverable	Status
Weekly progress report prepared	Completed
Intern name updated to Gayatri	Completed
Week 03 identified	Completed
PDF format	Completed

File name: Gayatri_Week-03.pdf	Ready for upload

Prepared by: Gayatri | **Internship:** Embedded Systems and IoT | **Week:** 03